CONVERSATION-STATE BASED APPROACH

Assignment on

Assignment Report submitted in partial fulfilment of the requirements for the Course of

DSA0308 – Natural Language Processing

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IN

ARTIFICIAL INTELLIGENCE AND Machine Learning

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CONVERSATION-STATE BASED APPROACH

1. Problem Statement and Problem Formulation

Problem Statement

An e-commerce chatbot may receive information about the same order over several conversational turns. A customer does not always repeat the complete information in every message. Instead, they may use expressions such as *it*, *this order*, *that item*, or *there* to refer to information already mentioned.

A chatbot that processes each message independently may lose this information and assign an incorrect intent. For example, in the given interaction, the customer first mentions a laptop order that has not arrived and later says:

“I don't want to cancel it. I just want to know when it will arrive.”

The meaning of **“it”** depends on the previous conversation. Similarly, the occurrence of *cancel* should not cause the chatbot to classify the message as Order Cancellation because the customer explicitly rejects cancellation.

The proposed solution therefore maintains a structured **conversation state** that is updated after every user message. This state stores entities, events, delivery status, temporal information, references, rejected actions and the current intent. The assignment requires conversational context and reference resolution to be handled explicitly.

Problem Formulation

Input:
Multiple customer messages belonging to one e-commerce conversation.

Internal Representation:

A continuously updated conversation state.

Output:

- Current intent
- Relevant entities
- Resolved references
- Conversation status
- Intent confidence

- Retrieved information
- Generated response
- Tamil translation

The possible intent classes are Order Tracking, Order Cancellation, Product Information, Return Request, Refund Request, Payment Problem, Delivery Information, Account Problem, Complaint and General Inquiry.

2. Objective and Expected Outcomes

Objectives

The main objective is to design a chatbot that understands the **current user intention using both the current message and previously stored conversation information**.

The specific objectives are:

1. Process each conversational turn.
2. Extract entities from every message.
3. Maintain an evolving conversation state.
4. Perform morphological processing.
5. Generate n-gram representations.
6. Perform POS tagging.
7. Analyze sentence structure using CFG.
8. Detect negated actions.
9. Build semantic representations.
10. Resolve references using previous turns.
11. Maintain entity chains.
12. Detect discourse relationships.
13. Determine the current intent from the complete state.
14. Retrieve information relevant to the intent.
15. Generate an appropriate response.
16. Translate the response into Tamil.

These objectives incorporate the course concepts specified in the assignment.

Expected Outcomes

For the given conversation, the state-based system should maintain:

Product = laptop

Order = existing customer order

Order Status = not received

Time = three days

Delivery Issue = delayed

Cancellation = rejected

Current Reference = existing order

Current Intent = order tracking

The final intent should therefore be:

Order Tracking / Delivery Information

3. Requirements, Constraints and Assumptions

Requirements

Functional Requirements

The chatbot must:

- accept multiple conversational turns;
- tokenize and normalize text;
- perform stemming;
- generate n-grams;
- perform POS tagging;
- construct syntactic representations;
- detect negation;
- perform semantic interpretation;
- identify entities;
- resolve references;
- maintain entity chains;

- analyze discourse;
- update the conversation state;
- determine the current intent;
- retrieve appropriate information;
- generate a response;
- translate the response.

The assignment specifically requires reference resolution for expressions such as *it*, *this* and *that*, along with conversational context maintenance.

State Requirements

The conversation state should be capable of storing:

```
ConversationState
├── customer
├── products
├── orders
├── order_status
├── delivery_status
├── temporal_information
├── references
├── negated_actions
├── discourse_relations
└── current_intent
```

Constraints

The system should not process every message as an isolated input.

It must:

- retain relevant previous information;
- resolve references from conversational context;
- explicitly identify negated actions;
- distinguish an entity from an intent;
- avoid replacing previously established facts without evidence;

- provide intermediate processing results;
- avoid generating responses inconsistent with the conversation.

These requirements are consistent with the technical constraints in the assignment.

Assumptions

1. Conversation turns arrive sequentially.
2. The system maintains state during a session.
3. The most recent compatible entity is generally preferred for resolving simple references.
4. Customer messages contain sufficient information for identifying the relevant intent.
5. A predefined collection of e-commerce intents exists.
6. The chatbot has access to a small information-retrieval store.
7. Translation is performed after response generation.

4. Application of Relevant Course Knowledge / Concepts

Unit I – Morphological Analysis

Each message is tokenized and stemmed before higher-level processing.

For example:

Word	**Stemmed Form**
ordered	order
arrived	arriv
delivered	deliv
cancelling	cancel
tracking	track

This allows different word forms to be associated with similar concepts.

Regular expressions can also identify structured information such as order numbers and dates.

Unit II – N-Grams and POS

N-grams are used to preserve local word relationships.

For:

“I don't want to cancel it.”

Possible features include:

Unigrams

want

cancel

Bigrams

want cancel

cancel it

Trigrams

don't want cancel

want cancel it

The n-gram containing the negation context is more informative than the isolated word *cancel*.

POS tagging is used to distinguish actions, objects and other grammatical components.

Unit III – CFG and Negation

A simplified grammar can represent the structure of a customer statement:

S → NP VP

VP → Aux VP

VP → V NP

NP → Det N

The system separately detects negation.

For:

“I don't want to cancel it.”

the state records:

Action = cancellation

Negated = TRUE

This prevents the cancellation action from becoming the current customer goal.

Unit IV – Semantic Analysis and WSD

The semantic representation describes events and relationships.

For example:

Customer ordered Laptop

Order has not arrived

Customer requests arrival information

Customer rejects cancellation

A simplified FOPC representation is:

Ordered(Customer, Laptop)

NotArrived(Order)

Wants(Customer, ArrivalInformation)

NotWants(Customer, Cancellation)

WSD helps interpret terms according to the e-commerce context.

For example, *order* is interpreted as a purchase transaction rather than another possible meaning.

Unit V – Discourse, Reference and Context

This is the **central component of Version 3**.

The system creates an entity chain:

laptop → order → it → existing order

The word **"it"** is therefore not interpreted independently.

Similarly:

- **but** → introduces contrast;
- **still** → indicates an unresolved condition;
- **just** → restricts the customer's requirement.

The assignment explicitly identifies reference resolution, entity chains, discourse relations and coherence as Unit V concepts.

5. Design / Proposed Solution / Methodology

The proposed architecture is organized around a **Conversation State Manager**.

Customer Turn

↓

Preprocessing

↓

Morphological Analysis

↓

N-Gram + POS Analysis

↓

CFG + Negation

↓

Semantic Analysis

↓

Entity Extraction

↓

Reference Resolution

↓

Discourse Analysis

↓

UPDATE CONVERSATION STATE

↓

Current Intent Identification

↓

Information Retrieval

↓

Response Generation

↓

Tamil Translation

↓

Next Conversation Turn

The state is updated after every user message.

Conversation State Example

After the first message:

“I ordered a laptop three days ago, but it still hasn't arrived.”

the state becomes:

```
ConversationState
├── product = laptop
├── order_exists = TRUE
├── order_age = three days
├── arrival_status = not_arrived
├── delivery_problem = TRUE
└── probable_intent = order_tracking
```

After the second message:

“I don't want to cancel it. I just want to know when it will arrive.”

the state is updated:

```
ConversationState
├── product = laptop
├── order_exists = TRUE
├── order_age = three days
├── arrival_status = not_arrived
├── cancellation = REJECTED
├── current_reference = order
├── requested_information = arrival_time
└── current_intent = order_tracking
```

This state representation is the major distinction of this solution.

6. Algorithm / Pseudocode

```
BEGIN

Initialize ConversationState

FOR each user_message:

    normalized_text ← preprocess(user_message)

    tokens ← tokenize(normalized_text)

    stems ← perform_stemming(tokens)

    ngrams ← generate_ngrams(tokens, 1, 3)

    pos_tags ← POS_tag(tokens)

    parse_tree ← CFG_parse(user_message)

    negation_info ← detect_negation(user_message)

    semantic_info ← semantic_analysis(user_message)

    entities ← extract_entities(user_message)

    references ← find_references(user_message)
```

```
FOR each reference:

    candidate_entities ← obtain_entities_from_state()

    resolved_entity ← select_best_contextual_match(
                      reference,
                      candidate_entities)

END FOR

discourse_info ← analyze_discourse(user_message)

update ConversationState with:

    entities
    resolved references
    temporal information
    order information
    delivery status
    negated actions
    discourse relations

candidate_intents ← generate_possible_intents(
                    current_message,
                    ConversationState)

FOR each candidate intent:

    calculate linguistic evidence
```

```
        calculate state/context evidence

        IF intent is explicitly negated:
            reduce intent score

        IF intent agrees with stored state:
            increase intent score

    END FOR

    current_intent ← highest_valid_score

    retrieved_data ← retrieve_information(
                    current_intent,
                    ConversationState)

    response ← generate_response(
            current_intent,
            retrieved_data)

    translated_response ← translate_to_tamil(response)

    display state
    display resolved references
    display intent
    display response

END FOR
```

END

The algorithm contains the functions, data structure, loops, conditions and scoring required by the assignment's algorithm-design expectations.

7. Implementation / Source Code and Environment / Tools Used

Environment

Component	**Tool**
Programming Language	Python
NLP Library	NLTK
Regex Processing	Python re
Tokenization	NLTK
Stemming	PorterStemmer
POS Tagging	NLTK
CFG Parsing	NLTK
State Management	Python dictionaries/classes
Intent Processing	Python
Information Retrieval	Dictionary-based knowledge base
Translation	Suitable translation library/API

Conversation State Implementation

A Python class can represent the state:

```
class ConversationState:

    def __init__(self):
        self.product = None
```

```
    self.order = None
    self.order_status = None
    self.delivery_status = None
    self.time = None
    self.references = {}
    self.negated_actions = []
    self.discourse = []
    self.intent = None
```

After every message, the state is updated rather than recreated.

For example:

```
state.product = "laptop"
state.order_status = "existing"
state.delivery_status = "not_received"
state.time = "three days"
state.negated_actions.append("cancel")
state.intent = "order_tracking"
```

The important feature is that the second message can access the information extracted from the first message.

8. Test Cases and Expected / Actual Results

Test Case	**Conversation**	**Expected Intent**
TC1	“Where is my order?”	Order Tracking
TC2	“Cancel my order.”	Order Cancellation
TC3	“My package hasn't arrived.”	Order Tracking
TC4	“Can I return this item?”	Return Request
TC5	“I paid but the payment failed.”	Payment Problem

Test Case	Conversation	Expected Intent
TC6	"I don't want to cancel it. When will it arrive?" after discussing an order	Order Tracking
TC7	"What is its delivery status?" after mentioning an order	Order Tracking

Multi-Turn Test

Turn 1

"I ordered a laptop three days ago."

State:

product = laptop

order = existing

time = three days

Turn 2

"It hasn't arrived yet."

State becomes:

product = laptop

order = existing

delivery_status = not_received

reference("it") = order

Turn 3

"I don't want to cancel it."

State becomes:

reference("it") = order

cancellation = rejected

Turn 4

"I just want to know when it will arrive."

Final state:

requested_information = arrival_time

intent = order_tracking

Thus, the system uses **all four turns** rather than interpreting the final message independently.

9. Execution Screenshots / Outputs

The actual implementation should capture screenshots for the following.

Screenshot 1 – Preprocessing/tokenization output.

```
1. PREPROCESSING / TOKENIZATION
----------------------------------------------------------------------
['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', '.', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', ',', 'but', 'the', 'ba
ttery', 'is', 'disappointing', '.', 'it', 'does', 'not', 'last', 'long', ',', 'and', 'i', 'am', 'unhappy', 'with', 'it', '.', 'however', ',', 'the', 'service', 'team', 'replaced', 'it'
, 'quickly', ',', 'which', 'was', 'impressive', '.', 'overall', ',', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']
```

User:

I ordered a laptop three days ago,

but it still hasn't arrived.

Screenshot 2 – N-gram and POS-tagging output screenshot.

```
3. UNIGRAMS
----------------------------------------------------------------------
[('i',), ('was',), ('expecting',), ('the',), ('phone',), ('to',), ('be',), ('amazing',), ('.',), ('the',), ('display',), ('is',), ('bright',), ('and',), ('the',), ('camera',), ('takes'
,), ('excellent',), ('pictures',), (',',), ('but',), ('the',), ('battery',), ('is',), ('disappointing',), ('.',), ('it',), ('does',), ('not',), ('last',), ('long',), (',',), ('and',),
('i',), ('am',), ('unhappy',), ('with',), ('it',), ('.',), ('however',), (',',), ('the',), ('service',), ('team',), ('replaced',), ('it',), ('quickly',), (',',), ('which',), ('was',),
('impressive',), ('.',), ('overall',), (',',), ('i',), ('am',), ('satisfied',), ('with',), ('the',), ('phone',), ('.',)]

4. BIGRAMS
----------------------------------------------------------------------
[('i', 'was'), ('was', 'expecting'), ('expecting', 'the'), ('the', 'phone'), ('phone', 'to'), ('to', 'be'), ('be', 'amazing'), ('amazing', '.'), ('.', 'the'), ('the', 'display'), ('dis
play', 'is'), ('is', 'bright'), ('bright', 'and'), ('and', 'the'), ('the', 'camera'), ('camera', 'takes'), ('takes', 'excellent'), ('excellent', 'pictures'), ('pictures', ','), (',', '
but'), ('but', 'the'), ('the', 'battery'), ('battery', 'is'), ('is', 'disappointing'), ('disappointing', '.'), ('.', 'it'), ('it', 'does'), ('does', 'not'), ('not', 'last'), ('last', '
long'), ('long', ','), (',', 'and'), ('and', 'i'), ('i', 'am'), ('am', 'unhappy'), ('unhappy', 'with'), ('with', 'it'), ('it', '.'), ('.', 'however'), ('however', ','), (',', 'the'), (
'the', 'service'), ('service', 'team'), ('team', 'replaced'), ('replaced', 'it'), ('it', 'quickly'), ('quickly', ','), (',', 'which'), ('which', 'was'), ('was', 'impressive'), ('impres
sive', '.'), ('.', 'overall'), ('overall', ','), (',', 'i'), ('i', 'am'), ('am', 'satisfied'), ('satisfied', 'with'), ('with', 'the'), ('the', 'phone'), ('phone', '.')]

5. TRIGRAMS
----------------------------------------------------------------------
[('i', 'was', 'expecting'), ('was', 'expecting', 'the'), ('expecting', 'the', 'phone'), ('the', 'phone', 'to'), ('phone', 'to', 'be'), ('to', 'be', 'amazing'), ('be', 'amazing', '.'),
('amazing', '.', 'the'), ('.', 'the', 'display'), ('the', 'display', 'is'), ('display', 'is', 'bright'), ('is', 'bright', 'and'), ('bright', 'and', 'the'), ('and', 'the', 'camera'), ('
the', 'camera', 'takes'), ('camera', 'takes', 'excellent'), ('takes', 'excellent', 'pictures'), ('excellent', 'pictures', ','), ('pictures', ',', 'but'), (',', 'but', 'the'), ('but', '
the', 'battery'), ('the', 'battery', 'is'), ('battery', 'is', 'disappointing'), ('is', 'disappointing', '.'), ('disappointing', '.', 'it'), ('.', 'it', 'does'), ('it', 'does', 'not'),
('does', 'not', 'last'), ('not', 'last', 'long'), ('last', 'long', ','), ('long', ',', 'and'), (',', 'and', 'i'), ('and', 'i', 'am'), ('i', 'am', 'unhappy'), ('am', 'unhappy', 'with'),
('unhappy', 'with', 'it'), ('with', 'it', '.'), ('it', '.', 'however'), ('.', 'however', ','), ('however', ',', 'the'), (',', 'the', 'service'), ('the', 'service', 'team'), ('service',
'team', 'replaced'), ('team', 'replaced', 'it'), ('replaced', 'it', 'quickly'), ('it', 'quickly', ','), ('quickly', ',', 'which'), (',', 'which', 'was'), ('which', 'was', 'impressive')
, ('was', 'impressive', '.'), ('impressive', '.', 'overall'), ('.', 'overall', ','), ('overall', ',', 'i'), (',', 'i', 'am'), ('i', 'am', 'satisfied'), ('am', 'satisfied', 'with'), ('s
atisfied', 'with', 'the'), ('with', 'the', 'phone'), ('the', 'phone', '.')]
```

Screenshot 3 – Reference Resolution

```
10. WORD SENSE DISAMBIGUATION
----------------------------------------------------------------------

Word: service
Possible senses: customer support, software service
Selected sense: customer support
Reason: The context mentions a service team replacing the phone.

Word: display
Possible senses: phone screen, public presentation
Selected sense: phone screen
Reason: The word occurs in the context of a phone.

Word: battery
Possible senses: phone battery, generic battery
Selected sense: phone battery
Reason: The review is explicitly about a phone.

11. REFERENCE RESOLUTION
----------------------------------------------------------------------
It in 'It does not last long' -> Battery
it in 'I am unhappy with it' -> Phone
it in 'replaced it quickly' -> Phone
which in 'which was impressive' -> Service/replacement action
```

Reference:

"it"

Resolved To:

Existing Order

Screenshot 4 – Negation Analysis

```
3. NEGATION DETECTION
------------------------
Trigger: not
Scope: not last long ,
```

Detected Action:

Cancellation

Negation:

TRUE

Interpretation:

Customer rejects cancellation

Screenshot 5 – Final State

Conversation State

Product : Laptop

Order Status : Existing

Delivery Status : Not Received

Cancellation : Rejected

Current Reference : Order

Requested Information: Arrival Time

Current Intent : Order Tracking

Screenshot 6 – Response

Final Intent:

Order Tracking

Response:

Please provide your order number so I can check

the current delivery status of your laptop.

Tamil Response:

```
20. TAMIL TRANSLATION
----------------------------------------------------------------------
இந்த மதிப்பாய்வு ஒட்டுமொத்தமாக நேர்மறையானதாக வகைப்படுத்தப்படுகிறது. திரை மற்றும் கேமரா குறித்து நேர்மறையான கருத்துகள் உள்ளன. அதே நேரத்தில், பேட்டரி ஏமாற்றமளிப்பதா
கவும் நீண்ட நேரம் நீடிக்கவில்லை என்றும் கூறப்படுவதால் அது எதிர்மறையான அம்சமாக உள்ளது. சேவை அணியினர் தொலைபேசியை விரைவாக மாற்றியதால் சேவை அனுபவம் நேர்மறை
யாக உள்ளது. இறுதியாக, பயனர் தொலைபேசியில் திருப்தியாக இருப்பதாகக் கூறுகிறார். எனவே, ஒட்டுமொத்த மதிப்பீடு நேர்மறையாகும்.
```

The assignment requires evidence of intermediate NLP representations, final classification and generated chatbot output.

10. Results and Validation

Morphological Analysis

Word	Processed Form
ordered	order
arrived	arriv
cancelling	cancel
delivery	deliver
tracking	track

Conversation-State Validation

State Component	Value	Source
Product	Laptop	First turn
Order	Existing	First turn
Time	Three days	First turn
Delivery Status	Not received	First/second turn
Reference	“it” → order	Previous context
Cancellation	Rejected	Third turn
Requested Information	Arrival time	Final turn
Intent	Order Tracking	Complete conversation

Intent Validation

The critical distinction is:

cancel

↓

Potential cancellation signal

don't want to cancel

↓

Cancellation rejected

want to know when it will arrive

↓

Delivery/Tracking requirement

Therefore:

Final Intent: Order Tracking / Delivery Information

The assignment specifically requires the analysis to explain why *"I don't want to cancel it"* must not lead to Order Cancellation.

Validation Cases

Linguistic Feature	**System Behaviour**
Morphological variation	Normalizes related forms
N-gram dependency	Considers surrounding words
POS information	Provides grammatical evidence
Negation	Rejects negated intent
Reference	Resolves "it" from state
Ambiguity	Uses conversational context
Discourse	Interprets "but", "still", "just"

Linguistic Feature	System Behaviour
Multi-turn context	Retains previous information
Intent	Selects current user goal
Response	Generated according to state

These correspond to the validation areas required in the assignment.

11. Analysis, Comparison, Trade-offs and Justification

The major advantage of this approach is that it does not treat every message as an independent classification problem.

Consider:

“I ordered a laptop three days ago.”

At this point:

Intent = possibly Order Tracking

Then:

“It hasn't arrived.”

The reference **“it”** becomes meaningful because the state already contains the order.

Finally:

“I don't want to cancel it.”

The system already knows what *it* represents.

Importance of Reference Resolution

Without state:

"it" → unknown

With state:

"it" → existing order

This makes the chatbot much more capable of handling natural conversational behaviour.

Why “cancel” Does Not Become the Intent

The system stores:

negated_actions = ["cancel"]

Therefore:

cancel ≠ requested_intent

Instead:

requested_information = arrival time

and consequently:

intent = order tracking

Comparison

Approach	Main Advantage	Main Limitation
Keyword-based	Simple implementation	Cannot preserve context
Rule-based scoring	Explainable	Requires many manually designed rules
TF-IDF classifier	Learns statistical patterns	Limited conversation memory
Conversation-state model	Strong multi-turn understanding	State management is more complex
Proposed state-based approach	Context + linguistic processing	Requires careful state updates

Trade-off

The main trade-off is **complexity versus conversational reliability**.

Maintaining state requires additional memory and processing, but it substantially improves the system's ability to understand references and previous information.

This is particularly valuable in e-commerce because customers often refer back to an order without repeating its complete details.

12. Broader Considerations / SDG Relevance

The assignment maps the application to **SDG 9 – Industry, Innovation and Infrastructure**, with automated customer-service systems identified as an industry-relevant application.

Privacy

Conversation state may contain customer information such as products, order details and delivery information. The state should be protected and deleted appropriately after the session.

Bias

Reference resolution may perform differently depending on sentence construction and language style. Testing should include varied conversational expressions.

Incorrect State Updates

A major risk of this approach is storing an incorrect entity or reference. Once incorrect information enters the state, later responses may also become incorrect.

Accessibility

Tamil translation allows the system to provide customer support in a regional language.

Responsible AI

The chatbot should not perform irreversible actions merely because an intent was inferred. Actions such as cancellation should require explicit confirmation.

13. Conclusion, Limitations and Possible Improvements

Conclusion

A conversation-state-based e-commerce chatbot was designed to address the limitations of independent message classification. The system maintains information from previous turns and uses this information to interpret references, negation, entities and discourse.

The conversation state stores details such as the product, order status, delivery condition, temporal information, references and rejected actions.

For the given example, **"it"** is resolved to the customer's existing order, while the cancellation request is identified as negated. The customer's actual requirement is to determine when the order will arrive. Therefore, the final intent is **Order Tracking / Delivery Information**.

This demonstrates that maintaining conversational state can significantly improve the interpretation of multi-turn e-commerce interactions.

Limitations

- Incorrect entity extraction can affect later turns.
- Very long conversations can produce large states.
- Ambiguous references may have multiple possible antecedents.
- State management requires additional implementation effort.
- Complex discourse relations may require more advanced NLP models.
- The approach may struggle when customers suddenly change topics.

Possible Improvements

Future development could include:

- Transformer-based dialogue-state tracking.

- Advanced coreference resolution.
- Long-context conversational models.
- Automatic state-confidence estimation.
- Multilingual conversation-state tracking.
- Integration with real-time order-management systems.
- Human-agent escalation for uncertain states.

14. Individual Contribution of Group Members

The following is an example structure and should be modified according to the **actual contributions**.

Member	Contribution
M Poojitha Reddy	**Problem formulation and system architecture** – Defined the e-commerce chatbot problem, objectives, inputs, outputs, and overall conversation-state architecture.
Ch Sai Sri	**Text preprocessing** – Performed tokenization, stop-word removal, normalization, stemming, and basic text cleaning of customer messages.
Sunaina	**POS tagging and linguistic analysis** – Implemented Part-of-Speech tagging and analyzed grammatical structures to support downstream NLP tasks.
Dhanyasree	**Entity extraction** – Identified and extracted entities such as order ID, product, delivery location, date, and order status from conversations.
Dhanasree	**Reference resolution and coreference handling** – Resolved expressions such as it, this order, that item, and there using previous conversational context.
Keerthana	**Conversation-state management** – Designed the structured conversation state to maintain entities, events, delivery status, temporal information, references, and previous conversation information across turns.
Lahari	**Negation and intent determination** – Handled negation and rejected actions, such as "I don't want to cancel it," and ensured the correct intent was identified despite misleading keywords.

Member	Contribution
Ramya Sri	**Discourse analysis and response generation** – Analyzed relationships between conversational turns and generated context-aware chatbot responses based on the resolved intent and conversation state.
Harshitha	**Tamil translation, testing and documentation** – Implemented Tamil translation support, tested different conversational scenarios, evaluated system performance, and prepared project documentation.

15. References

The final report should cite the resources actually used during development. The assignment requires appropriate references for textbooks, research papers, NLP libraries, datasets, software/tools, websites and AI-assisted tools where applicable.

Suggested categories include:

1. NLP course textbook.
2. NLTK documentation.
3. Python documentation.
4. Research literature on dialogue-state tracking and conversational NLP.
5. Documentation for the selected translation library/API.
6. Dataset documentation, if a dataset is used.
7. Software and development tools used during implementation.
8. AI-assisted tools, if actually used.

16. One-Page Individual Reflection

Individual Reflection

This assignment helped me understand that a conversational chatbot cannot always identify a user's intention from the current sentence alone. In real conversations, users often refer to information mentioned in earlier messages. Therefore, an effective chatbot needs to retain and use previous conversational information. This understanding led me to design the system using a structured conversation-state model.

One of the key design decisions was to maintain a structured state that stores entities, order details, delivery status, temporal information, references, negated actions, and the current intent. Rather than treating every message as an independent input, the system continuously updates this state as new messages arrive. This made it easier to understand the relationship between different turns in the conversation.

Reference resolution was one of the NLP components that had the greatest impact on determining the final intent. For example, the word “it” cannot be understood accurately without considering the previous conversation. By resolving “it” to the previously mentioned order, the system can correctly interpret the following message. Similarly, the statement “I don't want to cancel it” is correctly identified as a rejection of cancellation instead of being mistakenly classified as a cancellation request.

Another major learning outcome was understanding how different NLP techniques work together. Morphological processing helps normalize words, n-grams capture nearby word relationships, POS tagging provides grammatical information, and semantic processing helps determine the intended meaning. These outputs can then be combined with discourse information and the current conversation state to produce a more accurate interpretation.

One challenge I encountered was determining how the conversation state should be updated when a new message is received. An incorrect update can influence the interpretation of subsequent messages. This helped me understand the importance of using confidence checks and handling ambiguous references carefully when managing conversation states.

The application also relates to **SDG 9 – Industry, Innovation and Infrastructure**, as conversational AI can contribute to automated digital customer-service systems. Supporting Tamil translation can further improve accessibility by allowing users who are more comfortable communicating in Tamil to interact with the system.

If I had more time to develop the project, I would improve the reference-resolution mechanism and implement a more advanced dialogue-state tracking approach. I would also evaluate the system using longer and more complex conversations, including situations where users switch topics or discuss multiple orders within the same conversation.

Overall, this assignment helped me move beyond learning individual NLP concepts and understand how they can work together in a practical conversational system. It strengthened my understanding of context management, algorithm design, linguistic analysis, and real-world NLP application development.